GuiA: Reinaixement Museum Interactive Audioguide

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This manuscript was compiled on April 22, 2026

1. Challenge Description

The main challenge addressed by this project is how to improve the visit to the Museum of the Renaixement in Molins de Rei, so that it becomes more understandable, engaging, and inclusive for different kinds of visitors.

The museum faces several challenges. One is communication: the museum must speak to segmented audiences with different interests, ages, levels of prior knowledge, and places of origin. Another challenge is gamification, or turning the visit into an interactive experience that keeps visitors active and curious without losing the seriousness of the cultural content. A device developed for these purposes needs to work with multiple information sources (website, social networks, the museum space itself, the artworks,..) so it can reflect the real identity of the institution rather than becoming an isolated digital tool.

In this context, we propose an AI adaptive audio guide as a technological solution to connect the audience with space, and history.

2. Challenge Context

The Museum of the Renaixement in Molins de Rei is a cultural space inside the Palau dels Requesens, a historic building that combines architecture, artworks, furniture, and everyday objects from the sixteenth century into the visit. This makes the museum especially rich as a heritage site, but also more demanding from a communication point of view, because visitors need to understand both the collection and the space in which it is displayed.

The museum receives around 5,800 visitors per year. As shown in Figure 1, visits are concentrated in the winter and spring months (January to May), with lower attendance during summer.

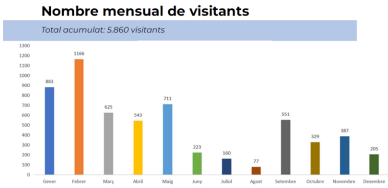

Fig. 1. Number of visitors by month. 2024 Report

Museum visitors do not share the same prior knowledge, age, or interests, and they may also have diverse accessibility needs - sensory, cognitive, or physical- which can influence how they navigate and interact with the museum. As shown in Figure 2, visitors also come from different locations with different native languages. The majority, in this case, come from Molins de Rei and the wider Barcelona and Catalonia area. But also a smaller proportion comes from the rest of Spain, Italy, and France, while a limited share comes from other parts of Europe and the rest of the world.

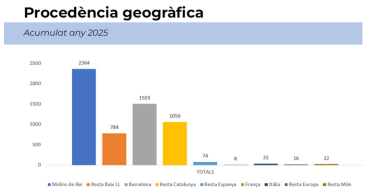

Fig. 2. Number of visitors by location. 2024 Report

Figure 3 shows that 1,095 visitors participated in guided visits in 2024, indicating that a significant portion of the public (18.68%) actively seeks mediated interpretation. Surveys conducted in Madrid (n>200) show 80% of the participants prefer institution-developed AI guides over generic solutions, valuing personalization and institutional identity GVAM (2025) However, it should be noted that the these guided visits and the audioguide currently provided by the museum are available exclusively in Catalan, which may limit accessibility for non-Catalan speaking visitors.

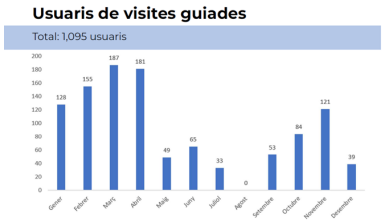

Fig. 3. Number of guided visits visitors. 2024 Report

Several commercial and institutional platforms currently offer digital museum guides. Smartify (2024) centers on artwork recognition through the smartphone camera, linking scanned objects to audio explanations. izi.TRAVEL (2024) allows museums and other institutions to build GPS-mapped audio routes. In the Spanish context, GVAM (2013) (Guías Virtuales Accesibles para Museos) is the dominant institutional model for interactive museum guides, deployed since 2013 in institutions such as the Museo Nacional del Prado or the Alhambra. GVAM (2013) guides offer multimedia

content, accessible routes, and personalised itineraries, but the content remains authored and fixed. MUVO (2023), a more recent Spanish commercial platform, advances toward adaptive delivery by tailoring content for different age groups using AI, although the extent to which it supports conversational interaction through large language models is not openly documented.

What distinguishes GuiA from all of these systems is its conversational and personalized large language model to produce responses based on what the visitor selects and asks in the moment. Visitors choose a guide language (English, Spanish, or Catalan) and a personality before entering, which shapes tone, register, and the way historical information is presented, and then indicate what they are looking at as they move through the museum. GuiA integrates spatial context (which room the visitor is in), object knowledge (artworks, furniture, architectural details), and history (sixteenth-century Catalonia, the Renaixement movement,..). The result is a dynamic explanation contextually grounded and visitor-specific. This adaptive model also generates interaction data — what visitors ask, which objects generate the most questions, which rooms produce confusion — that the museum can use to improve.

The Renaixement Museum operates within a layered governance structure: the Ajuntament de Molins de Rei holds ownership responsibility and funding authority; Molivers SL, the municipal public company owned by the Ajuntament, manages day-to-day operations; and the Servei Museus Molins, a municipal service established in 2022, provides the regulatory framework for the museums in Molins de Rei. The museum Direction holds authority over contenty. So the AI-mediated content must be aligned with this, and the knowledge graph that powers GuiA draws its content from the museum's own data.

On the legal side, the Web Content Accessibility Guidelines (WCAG), developed by the World Wide Web Consortium (W3C), establish technical standards for making digital interfaces perceivable, operable, understandable, and robust for people with disabilities. In Spain, compliance is legally required under Real Decreto 1112/2018, which transposes the European Accessibility Directive (EU 2016/2102) into national law, mandating WCAG 2.1 Level AA for public sector digital services. Because the Renaixement Museum is managed under municipal authority, its digital tools fall within the scope of this obligation.

Beyond accessibility, the General Data Protection Regulation (GDPR) applies to any personal data collected during interactions, including conversation logs and language preferences. The EU AI Act, adopted in 2024 and entering phased enforcement, additionally requires that visitors be informed they are interacting with an AI system, and that any profiling be proportionate and justified.

These conditions shape GuiA's design as a situated AI system, responsive to the museum's specific collection, audiences, and regulatory framework.

3. Stakeholder Mapping

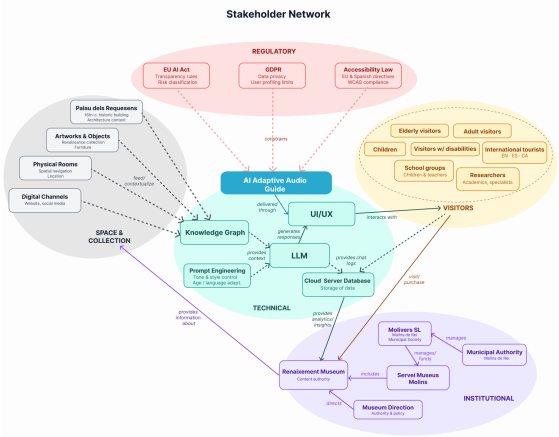

Fig. 4. Satakeholder Network showing relationships between the different actors that are present in the environment in which the AI device will be displayed

4. Real Life setting proposal

The most suitable real-life setting to test this device is a living lab inside the Museu del Renaixement itself, with a small set of visitors. According to the European Commission, living labs are experimentation spaces where innovative solutions are co-created and tested in a near-real-world environment with users, making them especially appropriate when the goal is to study both technical performance and user experience Kert et al. (2022). In this case, the museum offers a controlled setting where the AI audio guide can be evaluated while visitors move through the exhibition, and interact with the artworks.Kert et al. (2022).

The testing protocol would follow a short living-lab cycle. Visitors would be recruited voluntarily and briefly onboarded before selecting their language and guide profile. During the visit, anonymised interaction logs would be collected. After the visit, a short exit questionnaire will capture feedback from users, enabling the assessment of both technical performance and user experience in the real exhibition environment.

This setting is particularly valuable because the device is both an information tool and a mediation tool. Testing it inside the museum would allow us to observe how different audiences use the guide in practice, and whether the personalization, multilingual support, and navigation functions actually improve understanding and engagement Kert et al. (2022). It would also make it possible to detect practical issues such as usability, accessibility, and the balance between digital interaction and the museum's heritage atmosphere, which are difficult to assess outside the real exhibition space Kert et al. (2022).

Within this living lab, evaluation would combine quantitative and qualitative metrics. Quantitative indicators would include type of audience, time spent using the audioguide, and the number and type of questions asked. These data would be complemented by a post-visit questionnaire assessing perceived understanding of the exhibition, accessibility, engagement, cognitive load, and overall satisfaction (target > 80%). Accessibility-related aspects would focus on clarity of audio output, comprehension of explanations, and overall ease of use.

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